

Automated Segmentation of Post-Treatment Glioma Sub-regions and Resection Cavities from Multi-Parametric MRI

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Abstract—Precise segmentation of post-treatment gliomas is essential for medical response. This study presents a deep learning pipeline for the BraTS 2024 Post-Treatment Glioma challenge using multi-modal MRI (T1n, T1c, T2w, FLAIR) to segment four tumor sub-regions: Non-Enhancing Core (NETC), Edema (SNFH), Enhancing Tumor (ET), and Resection Cavity (RC). To address these post-surgical morphological complexities, we performed exploratory data analysis (EDA) and prepared a custom 3D multimodal data loader. We chose a baseline using a foundational 3D U-Net and subsequently transitioned to a Segmented Residual Network (SegResNet).

Index Terms—Brain Tumor Segmentation, BraTS 2024, Deep Learning, 3D U-Net, SegResNet, Multi-modal MRI, Post-Treatment Glioma, Medical Image Analysis

I. INTRODUCTION

Gliomas are brain tumors that need serious treatment, such as surgery and a combination of chemotherapy and radiation. After the treatment, it also needs to monitor the patient with MRI scans to see if the tumor is growing back or if there is a change in the tissue. Looking at these scans and detecting tumors can be very challenging and time-consuming, so we need computers to help with this task.

The BraTS 2024 Post-Treatment Glioma Challenge is a benchmark for evaluating how well a model performs on this task. It uses MRI images, such as T1n, T1c, T2w, and FLAIR, to examine four parts of the tumor: the Non-Enhancing Tumor Core, the Edema, the Enhancing Tumor, and the Resection Cavity. What makes this hard is that the brain images after treatment are very distorted, with some parts much larger than others, which makes it tough for computers to get it right.

This paper is about how we used learning to tackle this problem. We started by reviewing the data and developing a specialized tool to load the MRI images. We first used a 3D U-Net to test how well it would perform and to ensure our data was ready for use. Then we will switch to an advanced Segmented Residual Network, which is better at handling the different sizes and shapes of the tumor and the surrounding tissue, especially after surgery.

II. METHODOLOGY

A. Dataset and Software Stack

We utilized the BraTS-GLI dataset that contains 1,350 cases. Each case includes four 3D MRI modalities : T1, post-contrast

T1c, T2, and FLAIR. These are mapped to five discrete annotations: 0: Background, 1: Non-Enhancing Tumor Core (NETC), 2: Surrounding Non-enhancing FLAIR Hyperintensity (SNFH), 3: Enhancing Tumor (ET), and 4: Resection Cavity (RC). We implement our project using PyTorch, and the Medical Open Network for AI (MONAI) library. These frameworks are specifically chosen by us for their efficient processing capabilities for high-dimensional NIfTI medical image formats.

B. Multi-Modal Pipeline

We process the data using a custom MONAI-based pipeline that combines the four MRI modalities into a unified 4-channel 3D input tensor and applies channel-wise non-zero intensity normalization to standardize for scanner variations. To address class imbalance and GPU memory, we crop the foreground to remove empty background voxels, followed by a targeted $96 \times 96 \times 96$ voxel patch extraction, ensuring balance between both healthy tissue and underrepresented tumor/cavity regions. Additionally, to prevent overfitting, 3D spatial augmentations are applied, which included random spatial flipping across anatomical axes and volumetric rotations.

C. Model Training

1) *Data Processing and Loading*: Our data pipeline is built using a custom utility to handle the structural complexities of the BraTS dataset. Our Multi-Modal dataloaders manage the BraTS dataset. The model processes 4-channel MRI volumes: T1, post-contrast T1c, T2, and FLAIR. The training is performed with a batch size of 2. To ensure data integrity, our DataLoader includes a visualization phase to verify that the images and labels are properly aligned before training begins. We split the data into a validation set of 270 patients to monitor generalized performance.

2) *Network Architecture*: First, we are using a 3D U-net that is specifically designed for multi-class volumetric segmentation. It follows an encoder-decoder structure with five levels using channel sizes of (16, 32, 64, 128, 256). Residual units were included within the network to improve feature learning and support better gradient flow. Batch normalization is also applied throughout the model to maintain training stability. The output consists of five channels that represent

five discrete annotations: 0: Background, 1: Non-Enhancing Tumor Core (NETC), 2: Surrounding Non-enhancing FLAIR Hyperintensity (SNFH), 3: Enhancing Tumor (ET), and 4: Resection Cavity (RC).

3) *Training Protocol*: The model was trained on a CUDA-enabled GPU using the Adam optimizer with a learning rate of 1×10^{-4} . We applied Dice Loss with softmax activation to exclude the background class to address class imbalance. The large 3D MRI volumes were efficiently processed during validation using sliding window inference with a ROI size of $96 \times 96 \times 96$. To prevent overfitting, we add early stopping criteria with a patience of 5 validation cycles (10 epochs). The best-performing model was saved based on the Global Dice score.

4) *Advanced Evaluation Metrics*: The model is assessing using both global and lesion-specific metrics to make the evaluation more clinically meaningful. Mean Dice Similarity Coefficient (DSC) and the 95th Percentile Hausdorff Distance (HD95) were calculated across the entire 3D volume to evaluate the overall performance. Additionally, a lesion-wise analysis was performed to calculate DSC and HD95 for each individual lesion. This provides us detailed understanding of how well the model handled multi-focal tumor structure.

III. PRELIMINARY ANALYSIS & DISCUSSION

A. Exploratory Data Analysis

1) *EDA 1: Volumetric Class Analysis (in cm^3)*: Based on the performed EDA, SNFH is the most common sub-region (mean volume: 49.24 cm^3), while NETC is the least common (mean volume: 1.69 cm^3), based on volumetric analysis of 1,350 patients. High variance is observed in both the ET and RC classes, suggesting significant patient-to-patient variation in tumour presentation.

2) *EDA 2: Multi-Modality Intensity Distribution (Sampled)*: The four MRI modalities (T1n, T1c, T2w, and T2f) have very different, non-standardized distribution profiles, according to the Kernel Density Estimation (KDE) of sampled voxel intensities.

3) *EDA 3: Z-Axis Slice-wise Tumor Bounding Box*: Tumours are mostly found in the central slices of the MRI volumes, which usually span a contiguous depth of about 60 to 80 slices, according to Z-axis spatial analysis.

4) *EDA 4: Multi-Focal Tumor Analysis*: According to 3D connected component analysis, 7.3% of patients have multi-focal pathology with up to five different lesions, whereas the majority of patients present with a single contiguous tumour mass.

5) *EDA 5: Per Tumor Sub-Region Composition*: Extreme class imbalance in tumour presentations is highlighted by per-patient compositional analysis. While the NETC is completely absent in 58.1% of patients and makes up only 1.7% of the average volume, SNFH is common and dominant (averaging 67.3% of total tumour volume). Additionally, an inverse relationship between the percentages of ET and RC is revealed by scatter plot analysis, that indicates that these features rarely co-dominate in the same scan.

6) *EDA 6: Tumor Centroid Location Heatmap*: Significant spatial heterogeneity in tumour localization is revealed by voxel-level frequency mapping across all 1,350 patients. The maximum frequency at any single voxel is only 24.0%, indicating the highly variable and patient-specific structural nature of brain tumour presentations, even though tumours tend to cluster within the central cerebral regions (peaking at axial slice $Z=107$).

7) *EDA 7: Per label Co-occurrence Matrix*: Label co-occurrence and conditional probability analysis reveal that SNFH is universally present across the patient cohort (100%), acting as a structural baseline. Additionally, conditional analysis highlights strong inter-dependencies; for example, if the rare NETC is present, then there is a 98.9% probability that ET is also present, which indicates typical progression patterns that the model must learn to navigate.

B. Discussion

When we set up our first baseline model, our tests showed two important computational dynamics in 3D multi-modal MRI segmentation:

- **Maintaining Spatial Consistency**: Initial experiments employing full-volume training succeeded by patch-based evaluation (e.g., $96 \times 96 \times 96$ sliding window) resulted in a significant variation in the network's acquired receptive field. This gap can be fixed by making sure that the training crop dimensions are exactly the same as the inference window.
- **Hardware Optimization**: Processing uncropped, 4-channel 3D volumes puts a lot of stress on GPU memory, which can cause Out-of-Memory (OOM) errors and stop training too soon.

By making these structural improvements, we have successfully stabilised our baseline framework, which will serve as a very reliable base for future full-scale training and testing.

IV. CONCLUSION

Using this BraTS dataset, we successfully developed an effective 3D pipeline for segmenting post treatment glioma. By performing EDA on 1350 patients, we discovered important issues like class imbalance and uneven shapes of surgical cavities. In order to solve this, we build a data loader using MONAI that normalizes intensities and applies 3D augmentations. We worked first on 3D U-Net that verified that data is correctly processed and aligned and further we are moving towards SegResNet model. This model will improve accuracy in detecting small tumor fragments and complex tissue changes by providing doctors with accurate volumetric reports for better monitoring of patients.

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